

Photocatalysis

Using light to drive chemical reactions

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Photocatalysis is the acceleration of a chemical reaction using light-absorbing catalysts, enabling processes like pollutant breakdown and clean fuel production using sunlight as the energy source.

- DEFINITION

The acceleration of a chemical reaction through the use of a catalyst that absorbs light to generate reactive species.

- WHY IT MATTERS

Photocatalysis offers a way to drive useful chemical reactions using abundant sunlight instead of fossil-fuel-derived energy, supporting cleaner energy and pollution control technologies.

- WORKING PRINCIPLE

INPUT / PROBLEM

CORE PROCESS

OUTCOME / APPLICATION

A photocatalyst absorbs light energy, exciting electrons that generate highly reactive species capable of breaking down pollutants or driving reactions like water splitting for hydrogen production.

- QUICK FACTS

- Titanium dioxide, the most common photocatalyst, is also used in sunscreen and white paint
- Photocatalysis is being explored as a way to produce 'solar fuels' like hydrogen directly from sunlight and water

- KEY TERMS

Photocatalyst

Band gap

Photoexcitation

Reactive oxygen species

- CORE CONCEPTS

- Semiconductor band gap
- Photoexcitation
- Reactive oxygen species generation
- Catalyst surface reactions

- APPLICATIONS

- Self-cleaning surface coatings
- Air and water pollutant breakdown
- Solar hydrogen fuel production
- Antibacterial surface treatments

- REAL-LIFE EXAMPLES

- Self-cleaning window coatings
- Air-purifying paint using titanium dioxide
- Solar-powered water purification

- ADVANTAGES

- Uses renewable sunlight as an energy source
- Can break down pollutants into harmless byproducts
- Supports clean hydrogen fuel production

- LIMITATIONS

- Many photocatalysts are only efficient under UV light, not full sunlight
- Reaction rates can be slow at scale
- Catalyst degradation over time reduces efficiency

- CAREER OPPORTUNITIES

Photocatalysis researcher, Renewable energy scientist, Materials scientist, Environmental technology developer

- INDUSTRY APPLICATIONS

Renewable energy, Environmental remediation, Construction coatings, Water treatment

- RESEARCH AREAS

Visible-light photocatalysts, Solar hydrogen production, Photocatalytic water treatment

- FUTURE SCOPE

Development of visible-light-active photocatalysts is expected to significantly expand practical solar-driven chemical applications.

- CURRENT TRENDS

Active research into novel semiconductor materials that work efficiently under visible sunlight rather than just UV light.

SUMMARY

Photocatalysis harnesses light to drive chemical reactions, offering a sustainable route to pollution cleanup and clean fuel production.